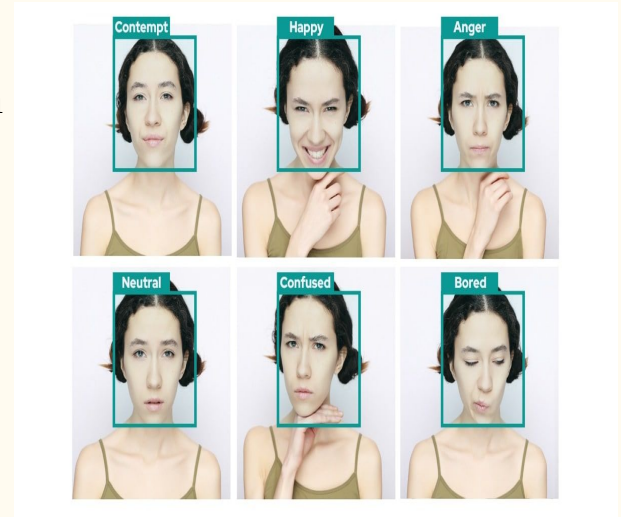


Emotion Detector

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Problem Statement

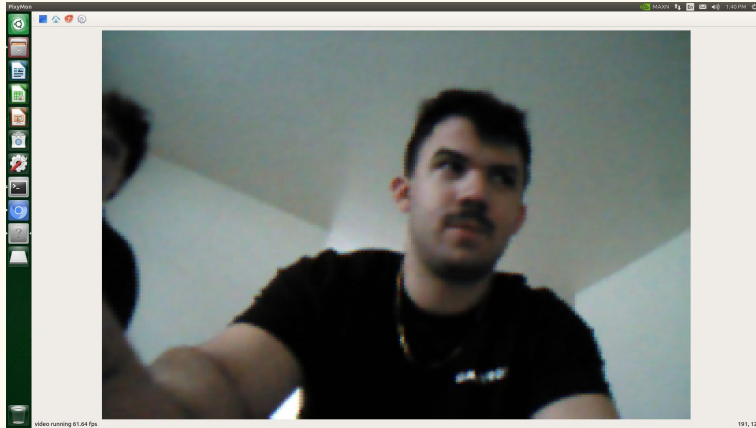
- Emotion recognition models are powerful but often rely on remote cloud servers to process video data
- Leads to latency, high bandwidth use, and privacy concerns
- Cloud dependence makes systems vulnerable to connectivity issues and limits real-world deployment
- Goal : Run emotion-detection fully on the edge using the NVIDIA Jetson, ensuring fast, private, and reliable results.



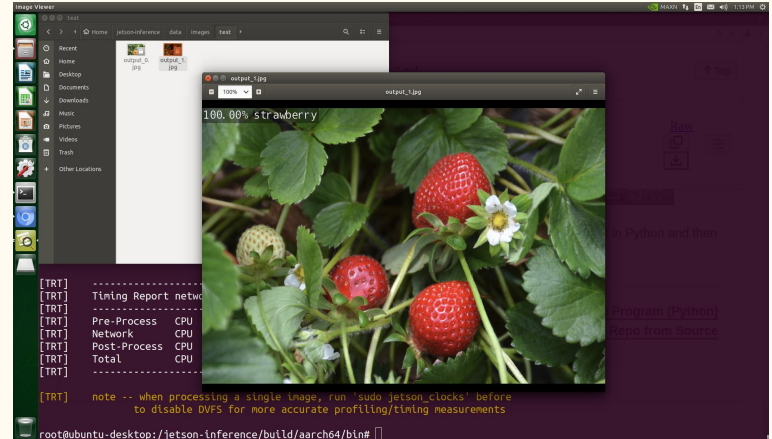
Progress

- Jetson setup complete
 - Installed NVIDIA JetPack SDK
 - Verified system stability and GPU acceleration.
- AI Inference Functional
 - Installed and ran Jetson-Inference (ImageNet) successfully
 - The jetson can now classify test images
- Webcam integration working
 - Connected and configured USB webcam for live video streaming.

Progress Pictures



- Webcam is connected and streaming live video directly to the Jetson.
- Demonstrates readiness for real-time emotion detection



- Jetson was able to caption an image of strawberries with 100% confidence.
- Confirms Jetson-inference and GPU acceleration work locally.

Future Work

- Train the emotion recognition model using the FER-2013 dataset.
- Deploy it on the Jetson and run real-time tests with webcam
- Optimize performance with TensorRT
- Display detected emotions on-screen or with LED feedback.

Q & A